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Wong

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(54) **VANITY MIRROR WITH HIDDEN SENSOR**

(71) Applicant: **simplehuman, LLC**, Torrance, CA (US)

(72) Inventor: **Ryan Wong**, Long Beach, CA (US)

(73) Assignee: **simplehuman, LLC**, Torrance, CA (US)

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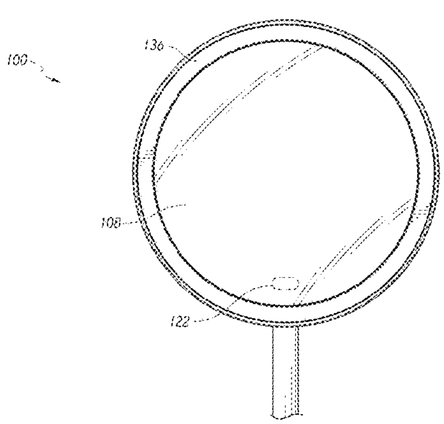
Primary Examiner — Bao Q Truong

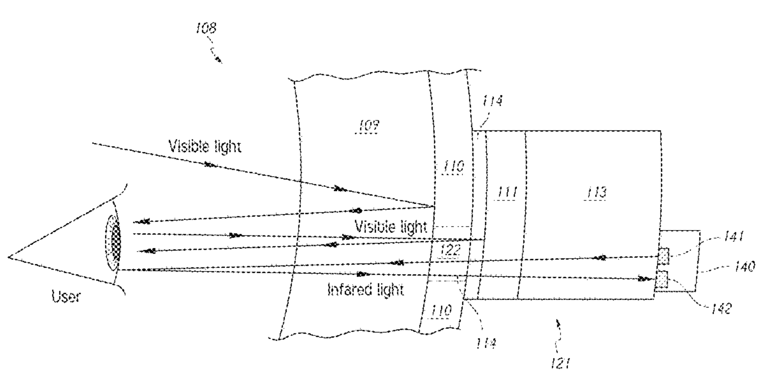
(74) *Attorney, Agent, or Firm* — Knobbe, Martens, Olson & Bear, LLP

(57) **ABSTRACT**

The present disclosure relates to a mirror assembly having a housing, a first mirror, a second mirror, a light source, and a sensor assembly. The first mirror can be positioned within an opening of the housing. The front surface of the second mirror can be coupled to a rear surface of the first mirror. The light source can be disposed on or at least partially around the first mirror. The sensor assembly can be positioned between a rear surface of the second mirror and the housing. The sensor assembly can include a transmitter positioned behind the rear surface of the second mirror and a receiver positioned behind the rear surface of the second mirror. The light source can be activated when the sensor assembly detects a user's presence.

23 Claims, 12 Drawing Sheets





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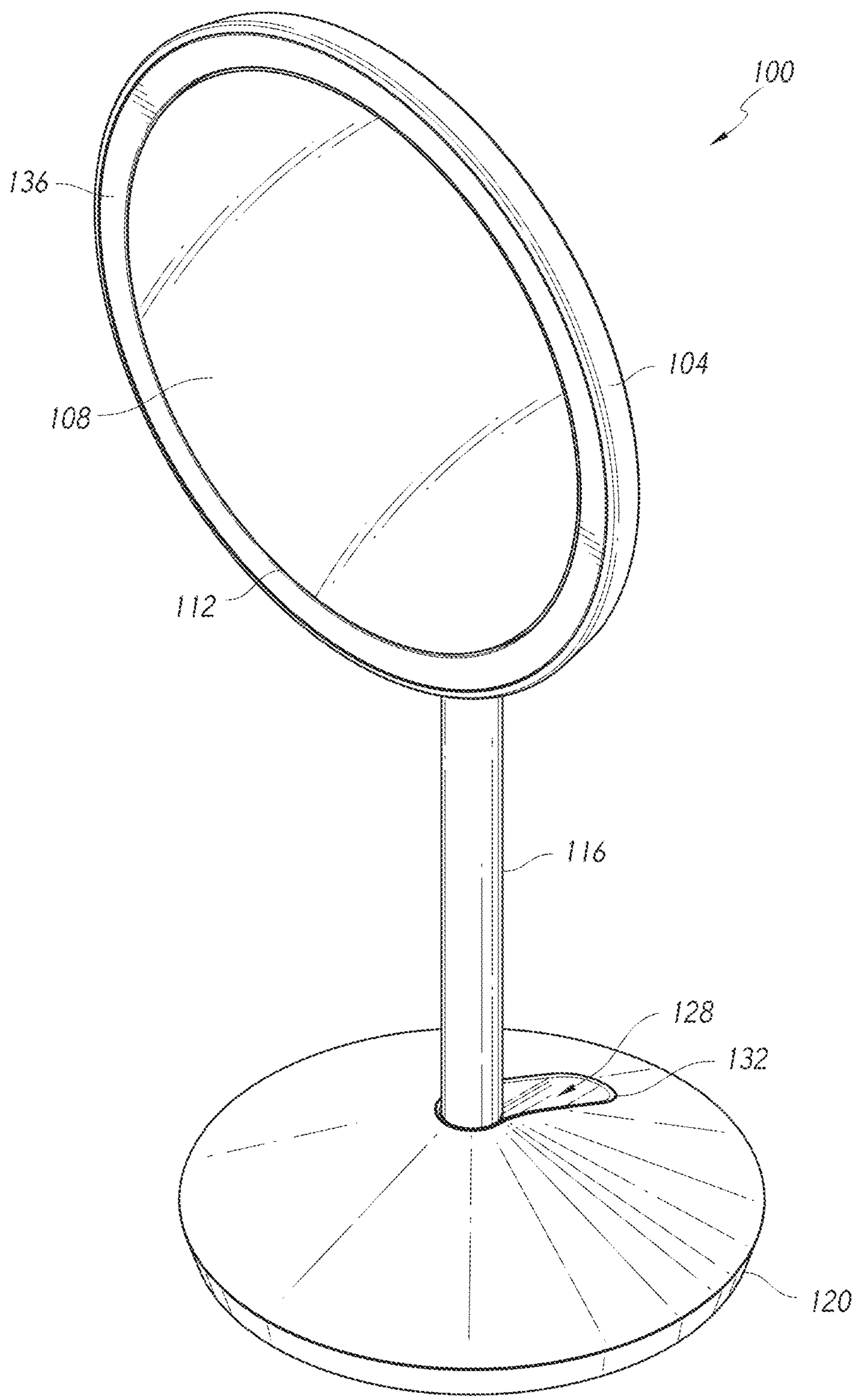

FIG. 1

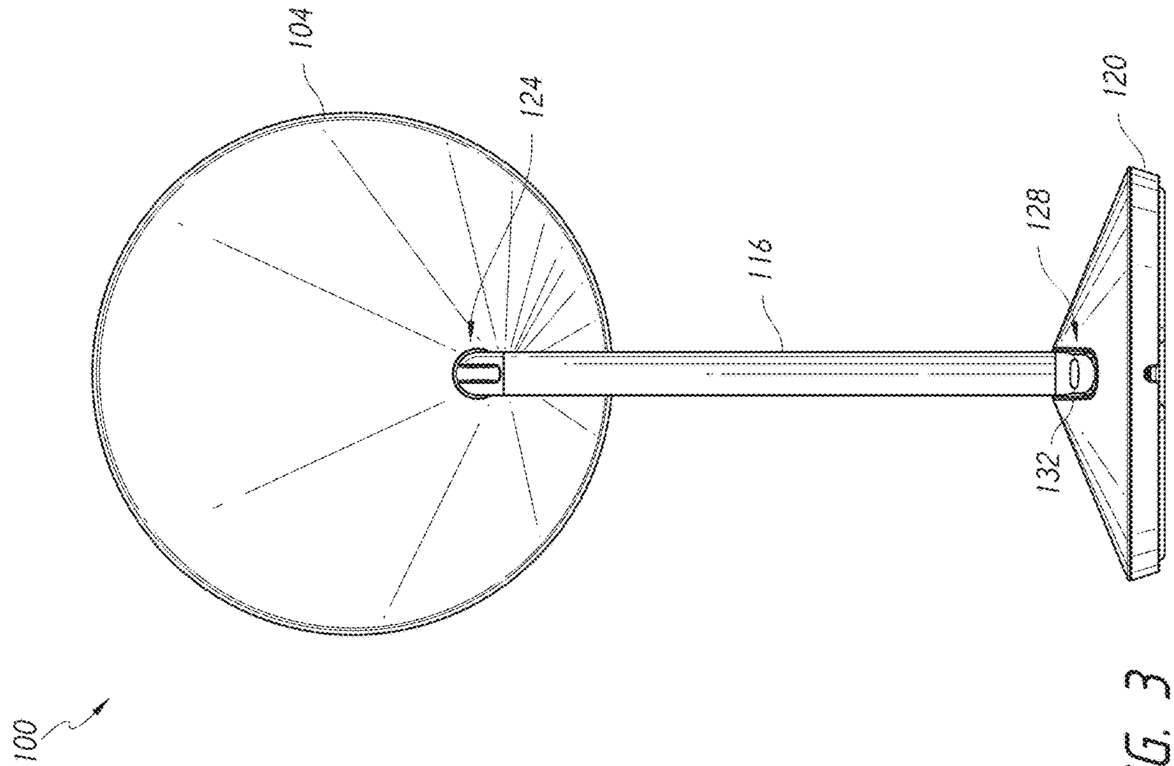

FIG. 3

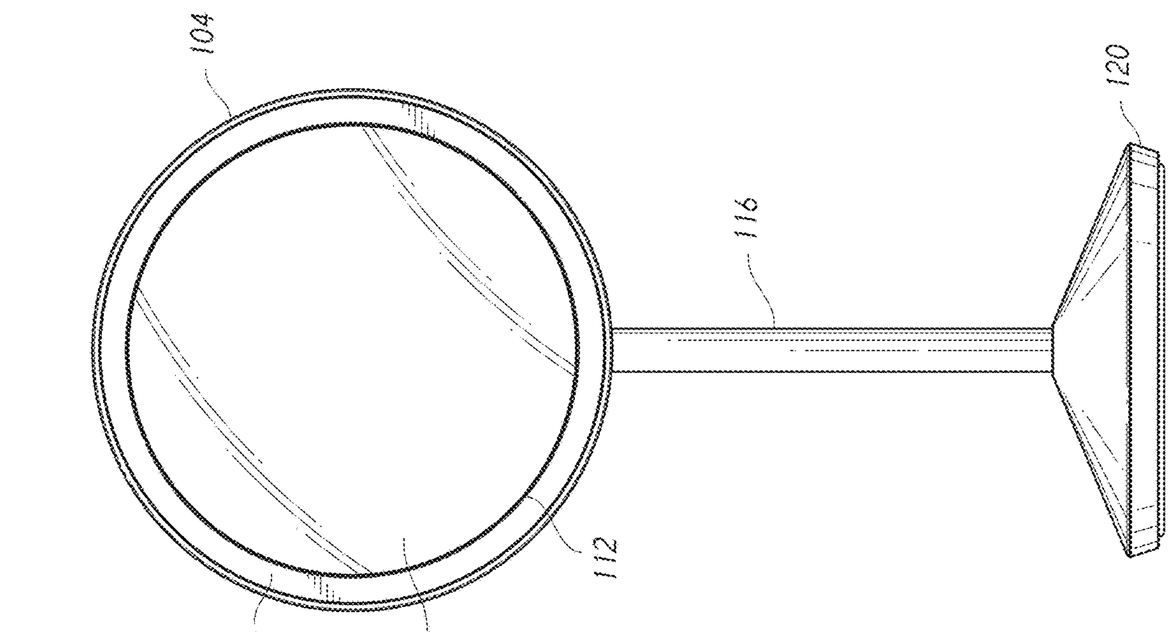

FIG. 2

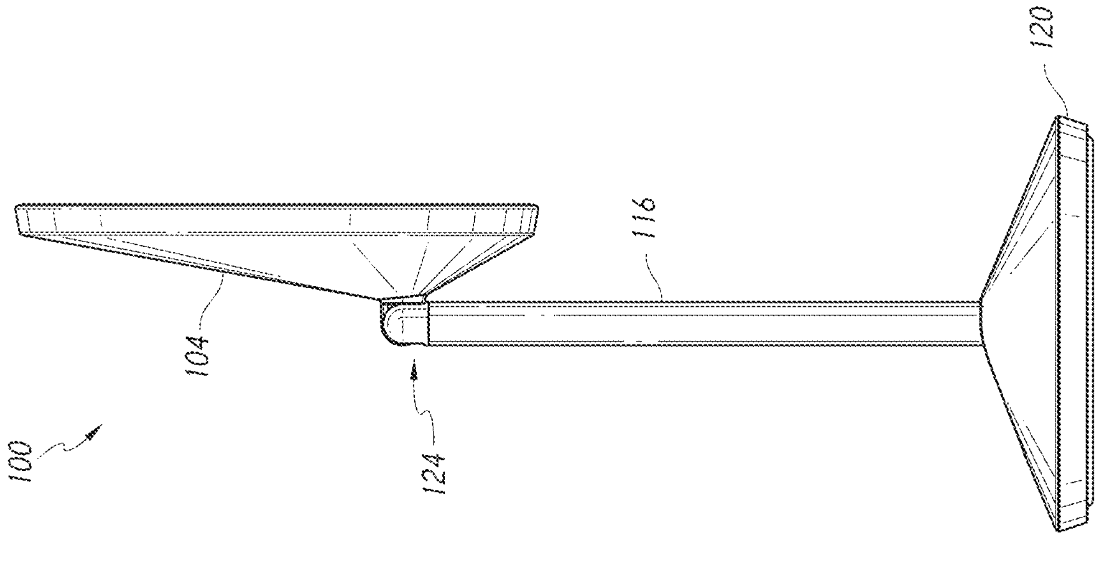

FIG. 5

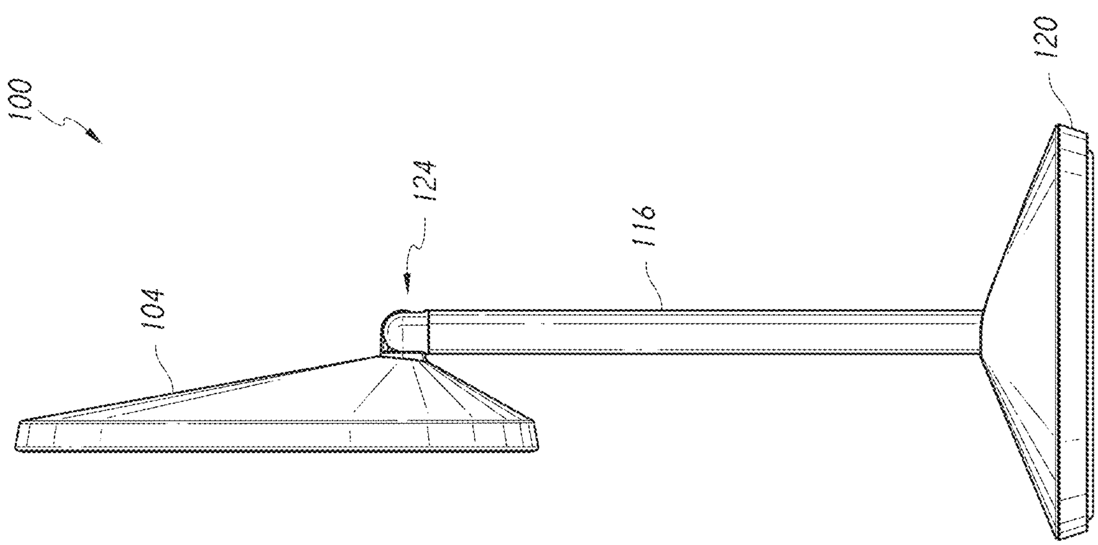

FIG. 4

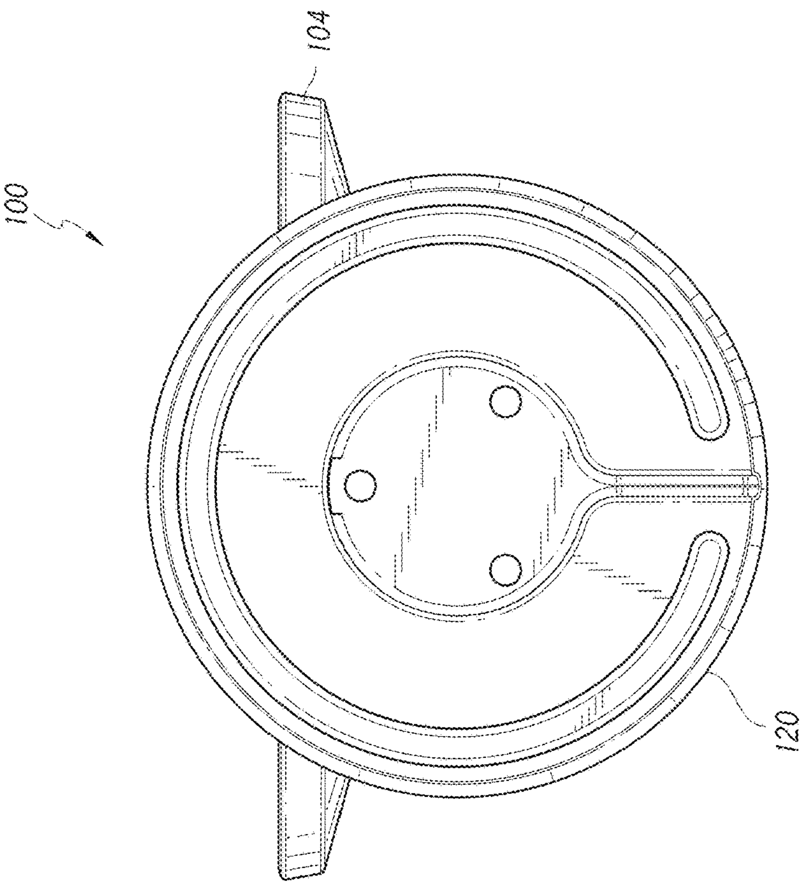

FIG. 7

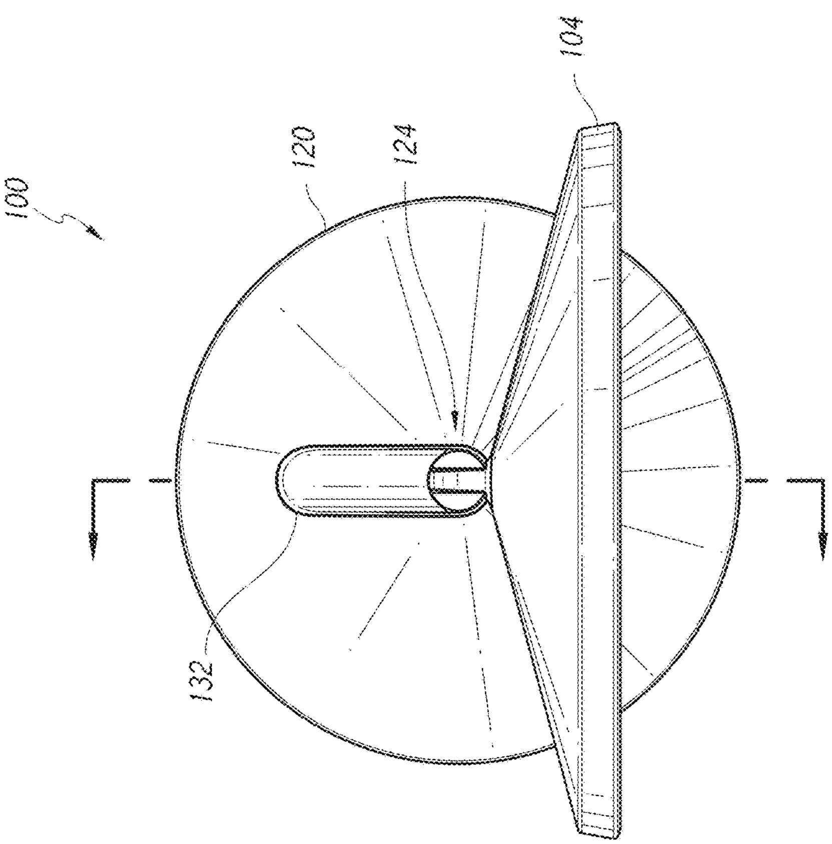

FIG. 6

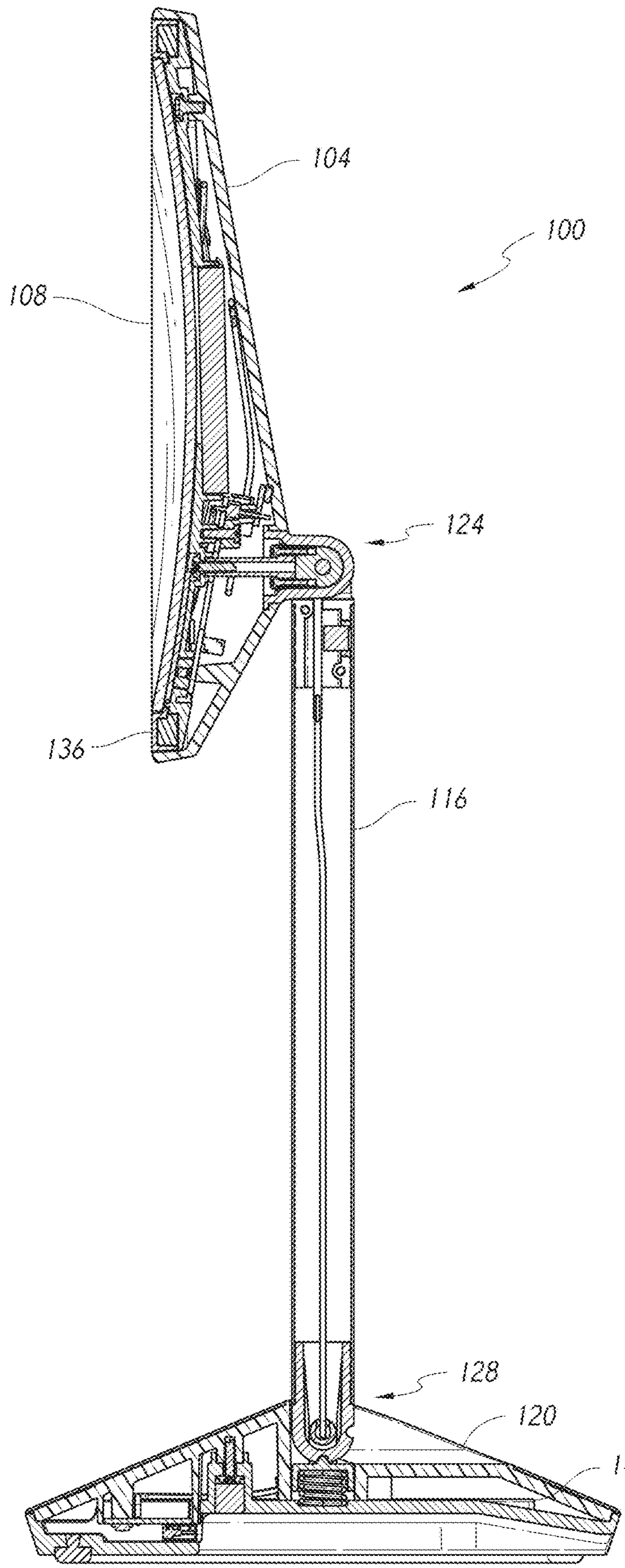

FIG. 8

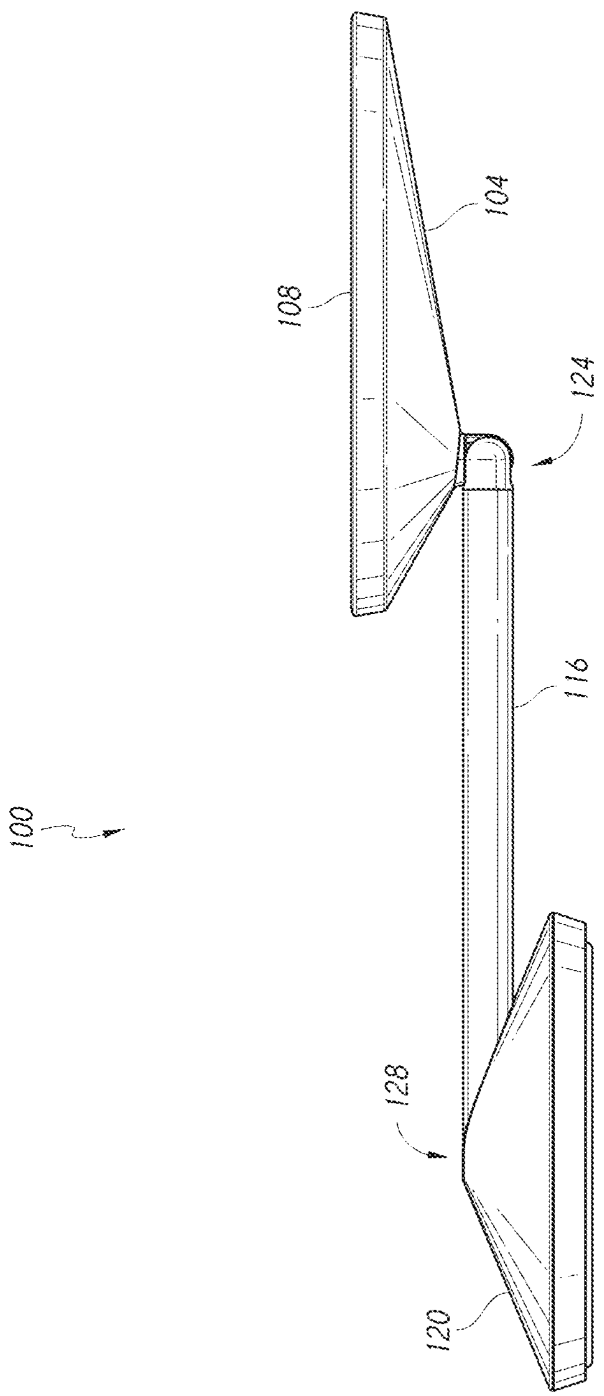

FIG. 9

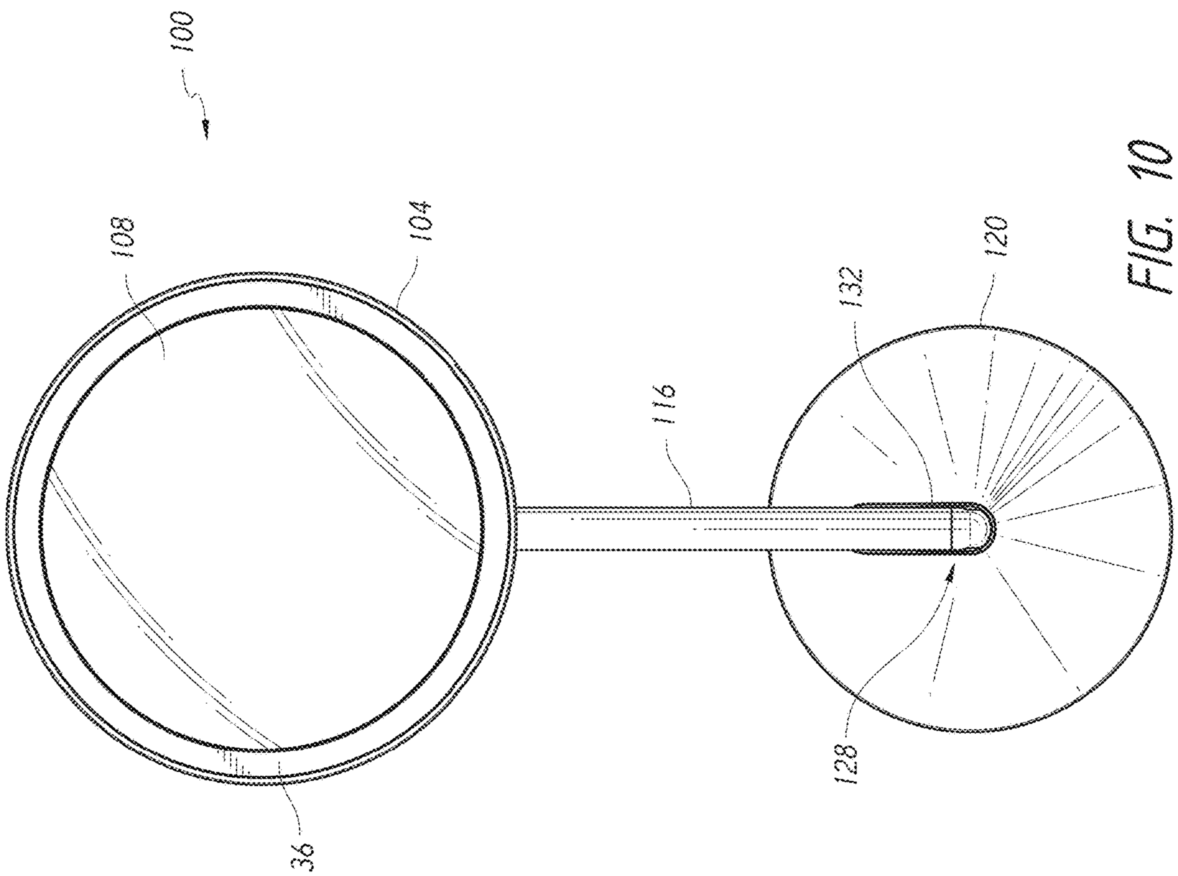

FIG. 10

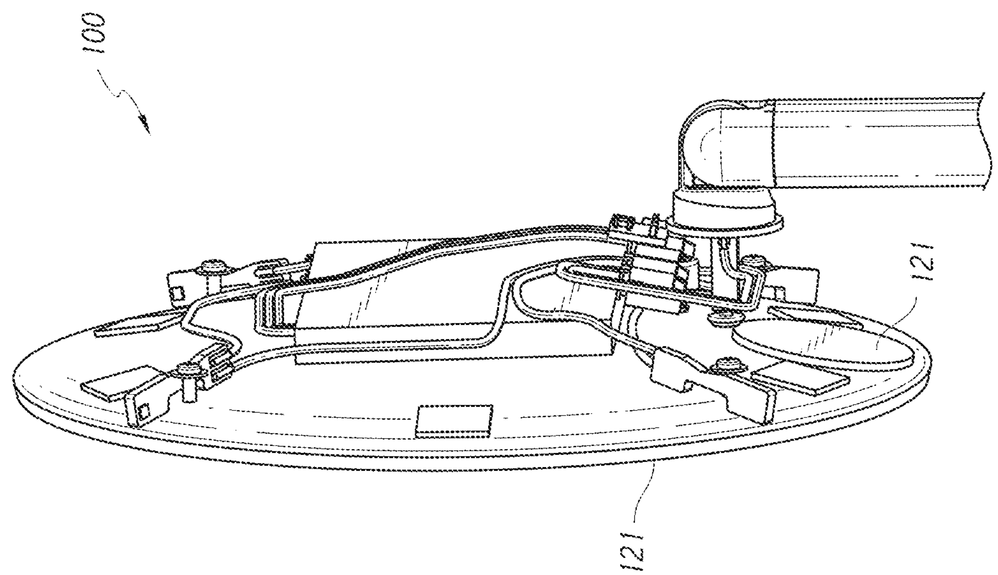

FIG. 12

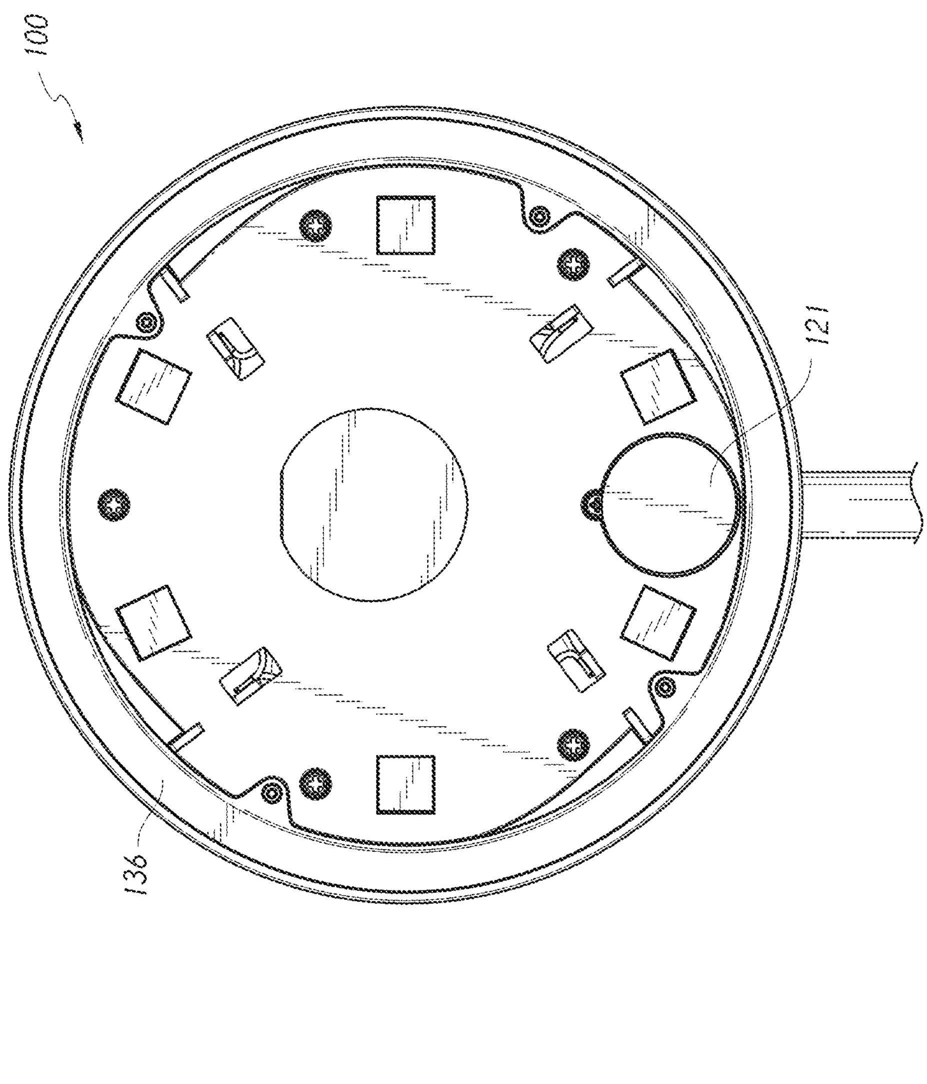

FIG. 11

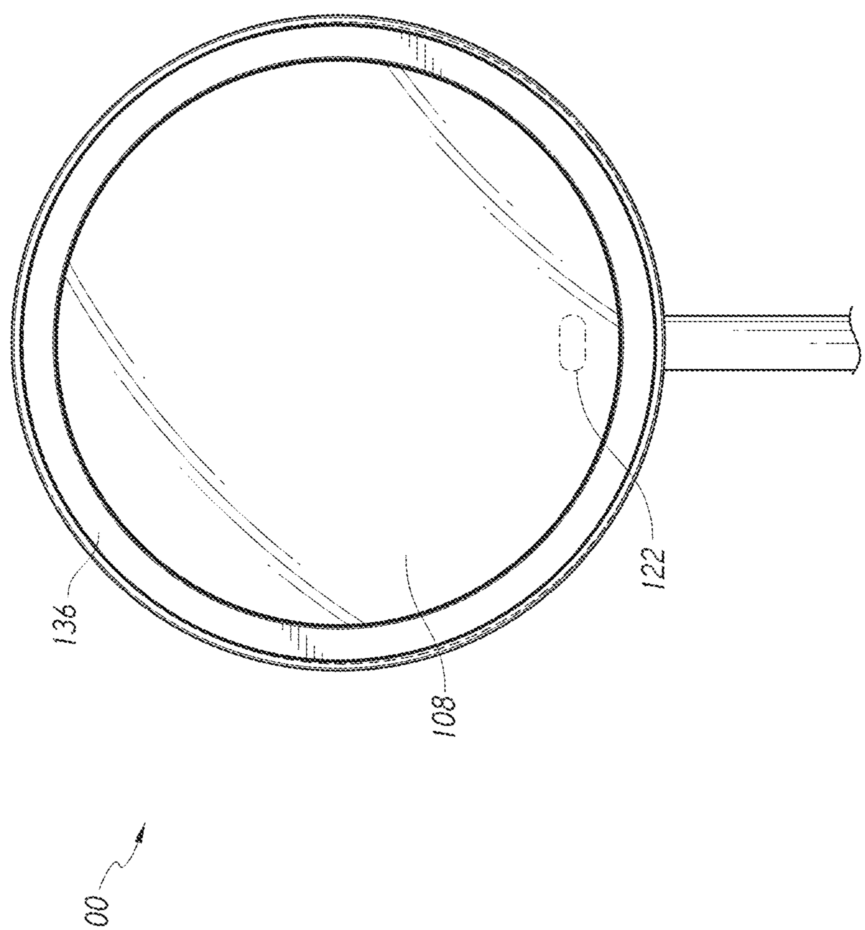

FIG. 13

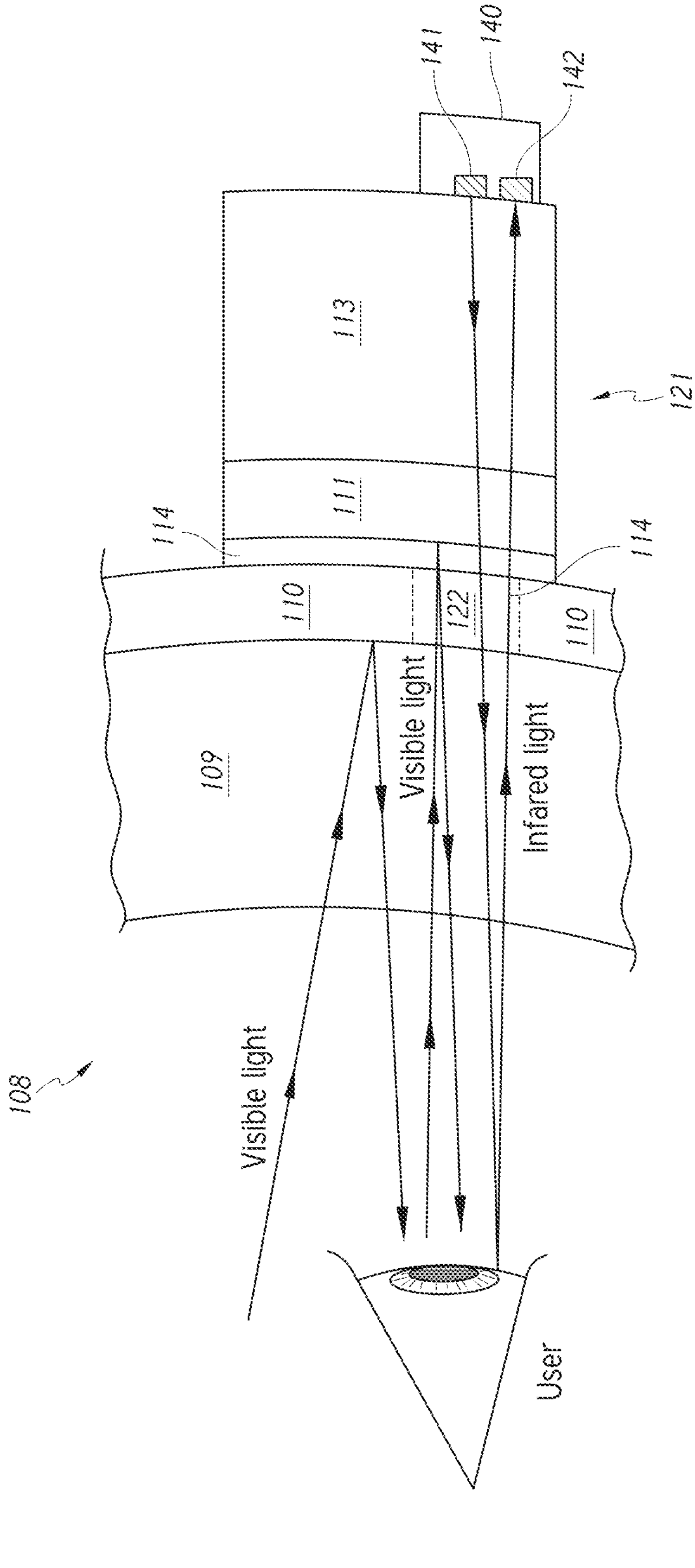
141
140
142
113
121
111
114
114
110
122
110
109
Visible light
Infared light
Visible light
108
User

FIG. 14

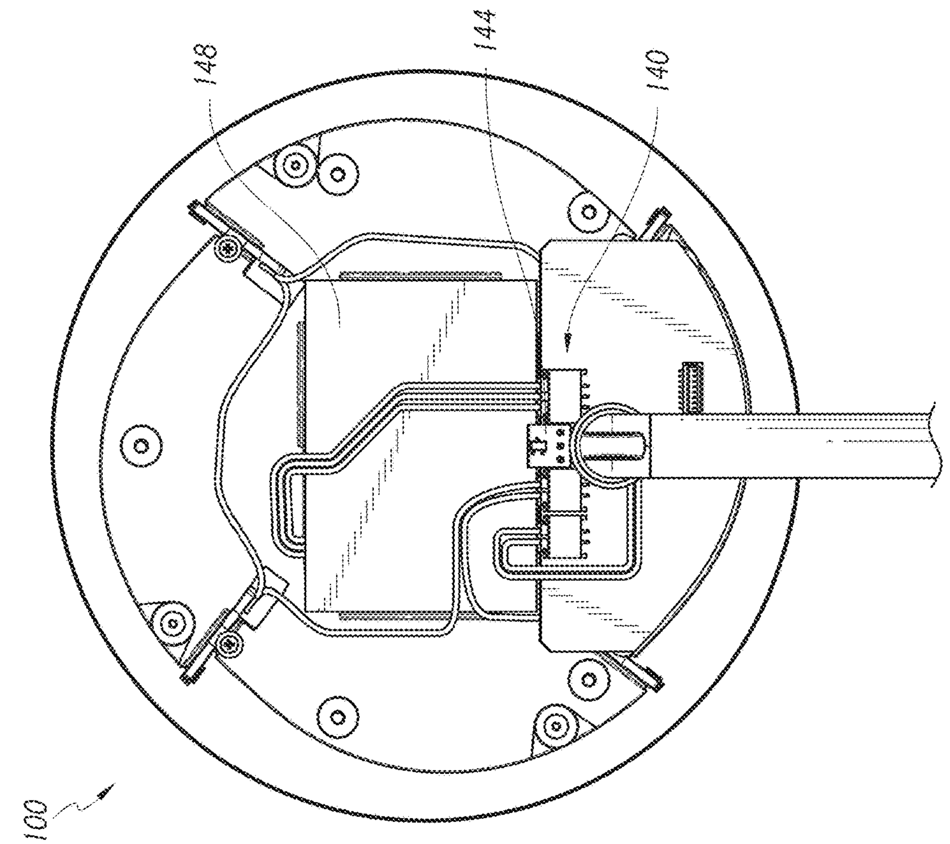

FIG. 16

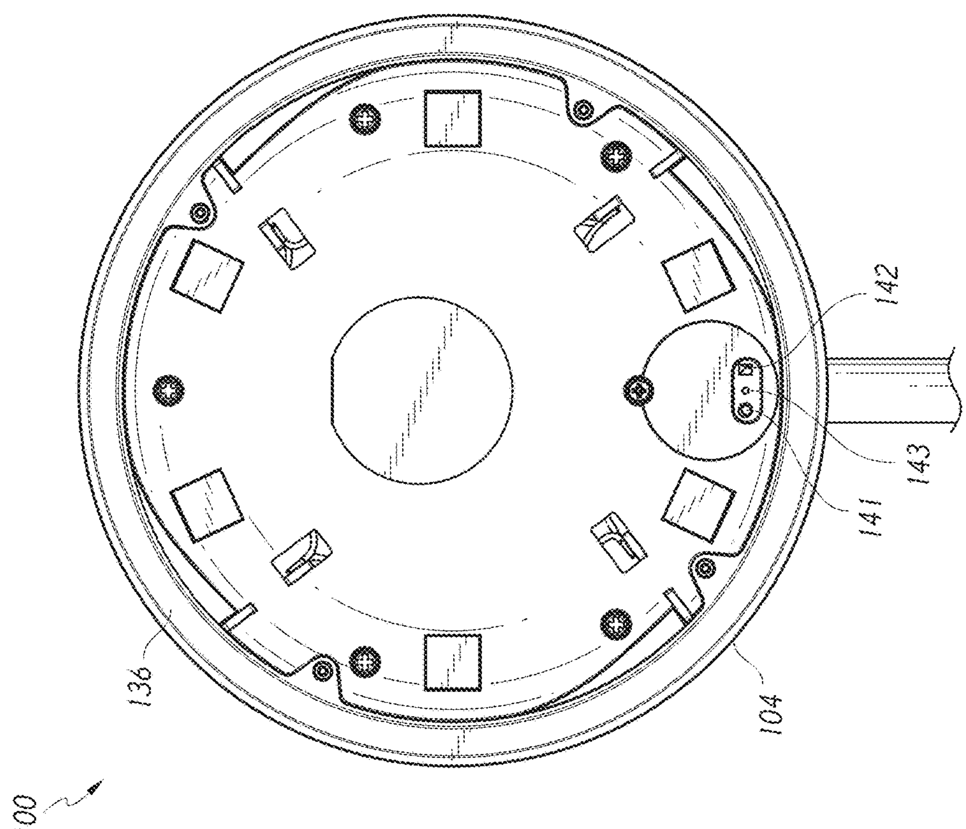

FIG. 15

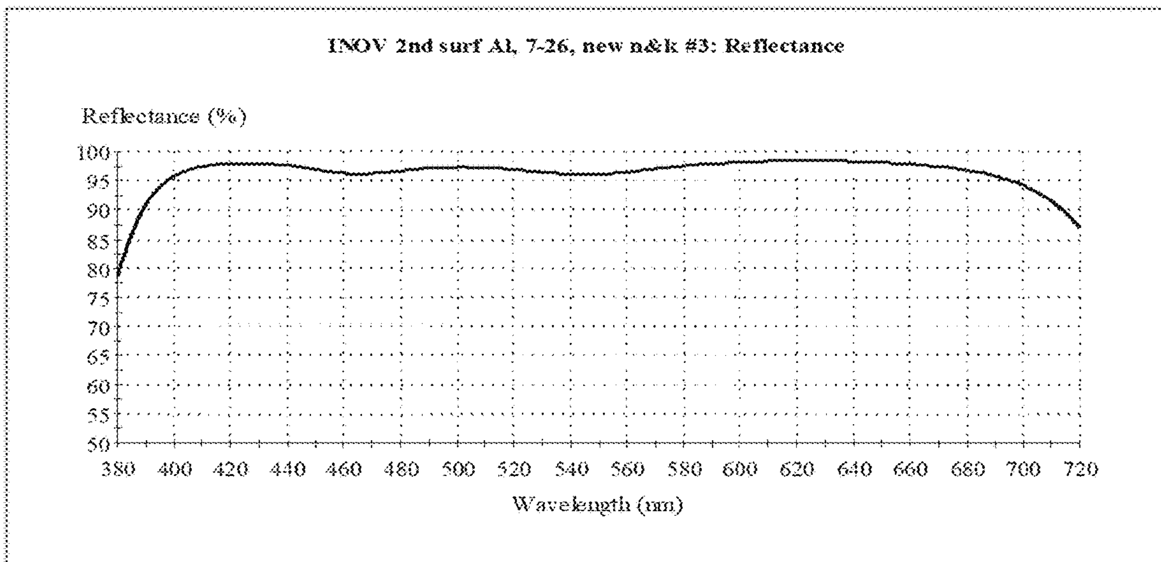
INOV 2nd surf Al, 7-26, new n&k #3: Reflectance
Reflectance (%)
100
95
90
85
80
75
70
65
60
55
50
380 400 420 440 460 480 500 520 540 560 580 600 620 640 660 680 700 720
Wavelength (nm)

FIGURE 17

VANITY MIRROR WITH HIDDEN SENSOR

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57. For example, this application claims the benefit of U.S. Patent Application No. 63/488,389 titled "VANITY MIRROR WITH HIDDEN SENSOR" and filed Mar. 3, 2023, the entirety of which is hereby incorporated by reference for all purposes and forms a part of this specification.

BACKGROUND

Field

The present disclosure relates to reflective devices such as mirrors, in particular to mirrors having lights that are activated when a user's presence is detected.

Description of the Related Art

Vanity mirrors are mirrors that are typically used for reflecting an image of a user during personal grooming, primping, cosmetic care, or the like. Vanity mirrors are available in different configurations, such as free-standing mirrors, hand-held mirrors, mirrors connected to vanity tables, bathroom wall mirrors, car mirrors, and/or mirrors attached to or produced by electronic screens or devices.

SUMMARY

The following disclosure describes non-limiting examples of some embodiments. For instance, other embodiments according to the present disclosure may or may not include any particular feature or any particular set of features described herein. Moreover, disclosed advantages and benefits may apply only to certain embodiments of the invention and should not be used to limit the disclosure.

In a mirror assembly that includes a proximity sensor for sensing the presence of a user and a controller for activating a user-illuminating light source when a user is sensed, the proximity sensor can be hidden from the user's view upon casual observation by the user. For example, the proximity sensor can be positioned behind or distal from the front, outer, or proximal-most reflective surface of the mirror assembly. The proximity sensor can be positioned on the opposite or back side of the outer or proximal-most reflective surface from the user. In some embodiments, the mirror assembly can hide the proximity sensor from the user by providing a front region of the mirror assembly that reflects a majority of visible light but that permits non-visible light or another form of sensing energy, such as acoustic waves, to pass through this region of the mirror assembly. The sensor positioned behind the front reflective surface can be configured to transmit sensing energy through a region of the mirror assembly toward the user and to receive the reflected energy back from the user, while permitting a majority of visible light that impinges on the front or outer surface of the region to be reflected back to the user.

In some embodiments, a mirror assembly includes a housing, a first mirror, a second mirror, a light source, and a sensor assembly. The housing includes an opening. The first mirror is positioned within the opening of the housing.

The first mirror has a front surface and a rear surface. The front surface is visible by a user. The second mirror has a front surface and a rear surface. Each of the first and second mirrors includes a thick substrate or layer of transparent material (e.g., glass or plastic). The front surface of the second mirror is coupled to the rear surface of the first mirror. The light source is disposed on or at least partially around the first mirror. The sensor assembly is positioned between a rear surface of the second mirror and the housing. The sensor assembly includes a transmitter positioned behind the rear surface of the second mirror and a receiver positioned behind the rear surface of the second mirror. The light source is activated when the sensor assembly detects a user's presence.

In some embodiments, the sensor assembly includes a proximity sensor. In some embodiments, the transmitter and the receiver are positioned behind a window of the first mirror. In some embodiments, the mirror assembly includes an indicator or display configured to convey a signal or information to a user through a reflective surface of the mirror assembly, such as an LED light, positioned behind the window of the first mirror. In some embodiments, the LED light can be configured to emit one form of light (e.g., yellow light and/or flashing light) when the mirror assembly is charging, another form of light (e.g., green light and/or flashing light) when the mirror assembly is finished charging, and/or another form of light (e.g., red light and/or flashing light) when the mirror assembly has a low-power or near-depleted battery. In some embodiments, the window is substantially invisible to or hidden from the user upon casual observation in normal use. In some embodiments, the second mirror is smaller than the first mirror. In some embodiments, the first mirror and the second mirror have the same curvature.

In some embodiments, the second mirror is coupled to the rear surface of the first mirror via an optical adhesive. In some embodiments, the optical adhesive is made of a material that mimics, or has substantially the same, index of refraction as the transparent material of the first mirror. In some embodiments, the first mirror has a layer or coating on the rear surface of the transparent material, and the second mirror has a layer or coating on the front surface of the transparent material. In some embodiments, the coating or layer of the first mirror comprises a reflective metal such as aluminum, and the layer or coating of the second mirror comprises a multilayer optical interference coating that includes a plurality of layers comprising two or more of layers comprising metal oxides and/or dielectrics.

In some embodiments, the mirror assembly includes a power source positioned within the housing. In some embodiments, the mirror assembly includes a shaft coupled at a first end to the housing and at a second end to a stand. In some embodiments, the shaft is rotatably coupled at the first end to the housing and the shaft is rotatably coupled at the second end to the stand.

In some embodiments, a mirror assembly includes first mirror, a second mirror, and a sensor assembly. The first mirror has a front surface and a rear surface. The front surface is visible by a user. The second mirror has a front surface and a rear surface. The front surface of the second mirror is coupled to the rear surface of the first mirror. The sensor assembly is positioned behind a rear surface of the second mirror. The sensor assembly includes a transmitter configured to emit light and a receiver configured to receive light. The sensor assembly is configured to detect a user's presence.

In some embodiments, the sensor assembly includes a proximity sensor. In some embodiments, the mirror assembly includes a light source, wherein when a user's presence is detected the light source is activated. In some embodiments, the transmitter and the receiver are positioned behind a window of the first mirror. In some embodiments, the second mirror is smaller than the first mirror. In some embodiments, the first mirror and the second mirror have the same curvature. In some embodiments, the second mirror is coupled to the rear surface of the first mirror via an adhesive. In some embodiments, the first mirror has a coating on the rear surface and the second mirror has a coating on the front surface. In some embodiments, the coating of the first mirror is aluminum and the coating of the second mirror is an optical interference coating.

BRIEF DESCRIPTION OF THE DRAWINGS

Some of these drawings are schematic, showing some examples of basic parts and concepts. Many different or additional structures, implementations, components, mechanisms, steps, and processes can be used. The claimed inventions should not be limited in any way to anything illustrated in the drawings.

FIG. **1** illustrates a front perspective view of an embodiment of a mirror assembly;

FIG. **2** illustrates a front view of the mirror assembly of FIG. **1**;

FIG. **3** illustrates a rear view of the mirror assembly of FIG. **1**;

FIG. **4** illustrates a right side view of the mirror assembly of FIG. **1**;

FIG. **5** illustrates a left side view of the mirror assembly of FIG. **1**;

FIG. **6** illustrates a top view of the mirror assembly of FIG. **1**;

FIG. **7** illustrates a bottom view of the mirror assembly of FIG. **1**;

FIG. **8** illustrates a side cross-sectional view of the mirror assembly of FIG. **1**;

FIG. **9** illustrates a sideview of the mirror assembly of FIG. **1** in a stowed configuration;

FIG. **10** illustrates a top view of the mirror assembly of FIG. **1** in a stowed configuration;

FIG. **11** illustrates a front view of the mirror assembly of FIG. **1** with a first mirror removed for illustrative purposes;

FIG. **12** illustrates a side perspective view of the mirror assembly of FIG. **1** with a housing removed for illustrative purposes;

FIG. **13** illustrates a front view of the mirror assembly of FIG. **1**;

FIG. **14** illustrates a schematic cross-sectional side view of a portion of the layers of the first mirror and the second mirror of the mirror assembly of FIG. **1** when coupled;

FIG. **15** illustrates a front view of the mirror assembly of FIG. **1** with the first mirror and the second mirror removed for illustrative purposes; and

FIG. **16** illustrates a rear view of the mirror assembly of FIG. **1** with a back portion of the housing removed for illustrative purposes.

FIG. **17** is a graph showing the relationship between reflectance and wavelength.

DETAILED DESCRIPTION

This specification provides textual descriptions and illustrations of many devices, components, assemblies, and subassemblies. Any structure, material, function, method, or step that is described and/or illustrated in one example can be used by itself or with or instead of any structure, material, function, method, or step that is described and/or illustrated in another example or used in this field. The text and drawings merely provide examples and should not be interpreted as limiting or exclusive. No feature disclosed in this application is considered critical or indispensable. The relative sizes and proportions of the components illustrated in the drawings form part of the supporting disclosure of this specification but should not be considered to limit any claim unless recited in such claim.

Certain embodiments of a mirror assembly are disclosed in the context of a portable, free-standing vanity mirror, as it has particular utility in this context. However, the various aspects of the present disclosure can be used in many other contexts as well, such as wall-mounted mirrors, mirrors mounted on articles of furniture, automobile vanity mirrors (e.g., mirrors located in sun-visors), and otherwise. None of the features described herein are essentially or indispensable. Any feature, structure, or step disclosed herein can be replaced with or combined with any other feature, structure, or step disclosed herein, or omitted.

Embodiments of the present disclosure relate to mirror assemblies. In particular, many embodiments of this disclosure relate to mirror assemblies having one or more light sources that are activated when a user's presence is detected. Some embodiments include one or more proximity sensors that are positioned behind a visual image reflective surface, such as a mirror. The positioning of the sensor(s) behind the mirror can provide an improved visual appearance of the mirror assembly because the sensor is hidden behind the mirror rather than visible on a user facing surface of the mirror assembly. In some implementations, this can provide the benefit of a simpler, more aesthetically pleasing appearance, and the benefit of permitting more light to be emitted more uniformly around the complete perimeter or circumference of the mirror, since the sensor does not occupy a portion of the light-emitting ring or periphery.

FIGS. **1**-**8** illustrate an example embodiment of a mirror assembly **100**. The mirror assembly **100** can include a housing **104** and a visual image reflective surface, such as a mirror **108**. The mirror **108** can be positioned within an opening **112** of the housing **104**. In some embodiments, the mirror assembly **100** can include a shaft **116** that can couple the housing **104** to a stand **120** or base portion. The shaft **116** and/or the stand **120** can support the housing **104**. The housing **104** can comprise plastic, stainless steel, aluminum, or other suitable materials.

In some embodiments the housing **104** can be pivotably coupled to a first end of the shaft **116** at a pivot **124**. The pivot **124** can allow the housing portion **104** and mirror **108** to be pivoted in one or more directions (e.g., up, down, right, left, and/or in any other direction). For example, the pivot **124** can include a ball joint, one or more hinges, or otherwise.

In some embodiments, the shaft **116** can be pivotably coupled to the stand **120** at a second end of the shaft **116** at a pivot **128**. The pivot **128** can allow the shaft **116** to be pivoted in one or more directions (e.g., up, down, right, left, and/or in any other direction). The shaft **116** can be pivoted to transition the mirror assembly **100** to a stowed position as shown in FIGS. **9** and **10**. The shaft **116** can be pivoted to rest in a recess **132** of the stand **120**.

The mirror **108** can include a generally flat, curved, or generally spherical surface, which can be convex or concave. The radius of curvature can depend on the desired optical power. In some embodiments, the radius of curvature can be at least about 15 inches and/or less than or equal to about 30 inches. The focal length can be half of the radius of curvature. For example, the focal length can be at least about 7.5 inches and/or less than or equal to about 15 inches. In some embodiments, the radius of curvature can be at least about 18 inches and/or less than or equal to about 24 inches. In some embodiments, the mirror **108** can include a radius of curvature of about 20 inches and a focal length of about 10 inches. In some embodiments, the mirror **108** is aspherical, which can facilitate customization of the focal points.

In some embodiments, the radius of curvature of the mirror **108** is controlled such that the magnification (optical power) of the object is at least about 2 times larger and/or less than or equal to about 7 times larger. In certain embodiments, the magnification of the object is about 5 times larger. In some embodiments, the mirror can have a radius of curvature of about 19 inches and/or about 7 times magnification. In some embodiments, the mirror can have a radius of curvature of about 24 inches and/or about 5 times magnification.

As shown, the portion of the mirror assembly **100** that faces the user is the mirror **108**, which can have a generally circular shape. In some embodiments, the mirror **108** can have an overall shape that is generally elliptical, generally square, generally rectangular, or any other shape. In some embodiments, the mirror **108** can have a diameter of at least about 8 inches and/or less than or equal to about 12 inches. In some embodiments, the mirror **108** can have a diameter of about 8 inches. In certain embodiments, the mirror **108** can have a diameter of at least about 12 inches and/or less than or equal to about 16 inches. In some embodiments, the mirror **108** can include a thickness of at least about 2 mm and/or less than or equal to about 3 mm. In some embodiments, the thickness is less than or equal to about two millimeters and/or greater than or equal to about three millimeters, depending on the desired properties of the mirror **108** (e.g., reduced weight or greater strength).

The mirror **108** can be highly reflective (e.g., has at least about 90% reflectivity). In some embodiments, the mirror **108** has greater than about 70% reflectivity and/or less than or equal to about 90% reflectivity. In other embodiments, the mirror **108** has at least about 80% reflectivity and/or less than or equal to about 100% reflectivity. In certain embodiments, the mirror **108** has about 87% reflectivity. The mirror **108** can be cut out or ground off from a larger mirror blank so that mirror edge distortions are diminished or eliminated. One or more filters can be provided on the mirror to adjust one or more parameters of the reflected light. In some embodiments, the filter comprises a film and/or a coating that absorbs or enhances the reflection of certain bandwidths of electromagnetic energy. In some embodiments, one or more color adjusting filters, such as a Makrolon filter, can be applied to the mirror to attenuate desired wavelengths of light in the visible spectrum.

The mirror **108** can be optical grade and/or comprise glass. For example, the mirror **108** can include ultra clear glass. Alternatively, the mirror **108** can include other transparent or translucent materials, such as crystal, plastic, nylon, acrylic, or other suitable materials. The mirror **108** can also include a coating or backing **110** that has high reflectivity and low transmissivity, such as a coating or backing **110** that includes a metal such as aluminum or silver. The coating **110** can be on a rear surface of the mirror **108** (e.g., the surface facing away from a user). In some embodiments, the coating **110** can impart a slightly colored tone, such as a slightly bluish tone to the mirror. In some embodiments, an aluminum coating can prevent rust formation and provide an even color tone. The mirror **108** can be manufactured using molding, machining, grinding, polishing, or other techniques.

The mirror assembly **100** can include one or more light sources **136** configured to transmit light. In some embodiments, the light source **136** can surround an outer perimeter of the mirror **108**, for example, the light source **136** can be a ring as shown. Light can be evenly or generally uniformly distributed around the entire perimeter of the mirror **108** without any optically significant breaks or interruptions in the light emitting perimeter (such as may otherwise occur if a proximity sensor is positioned along or near the perimeter of the mirror assembly **100**). In some embodiments, one or more light sources **136** can be positioned on or around a user facing side of the mirror assembly **100**. In some embodiments, one or more light sources **136** can be positioned on a side of the housing **104**. Various light sources **136** can be used. For example, the light sources **30** can include light emitting diodes (LEDs), fluorescent light sources, incandescent light sources, halogen light sources, or otherwise.

FIG. **11** illustrates the mirror assembly **100** with the first mirror **108** removed for illustrative purposes. FIG. **12** illustrates the mirror assembly **100** with the housing **104** removed for illustrative purposes. As shown in FIGS. **11** and **12**, a second mirror **121** can be positioned behind and/or coupled directly or indirectly to the first mirror **108**. In some embodiments, the first mirror **108** and the second mirror **121** can both be curved or non-planar as shown, such as with corresponding curvatures that permit the first mirror **108** and second mirrors **121** to fit closely together without any functionally significant gap between them. A rear surface of the first mirror **108** can be coupled with a front surface of the second mirror **121**. In some embodiments, the first mirror **108** and the second mirror **121** can be placed adjacent to each other or in contact with each other when the mirror assembly **100** is assembled, such as with a mechanical clip or bracket, or with optical adhesive **114**. In some embodiments, the optical adhesive **114** can be made of a material that has the same or substantially the same index of refraction as the transparent material **109** of the first mirror **108** so as to resist refraction of light at the boundary between the transparent material **109** and the optical adhesive **114**. The optical adhesive **114** can be used to couple a front surface **111** of the second mirror **121** to a rear surface **110** of the first mirror **108** and/or to fill any gap or avoid that may otherwise occur between or near them.

In some embodiments, the second mirror **121** can be smaller than the first mirror **108**. In some embodiments, the first mirror **108** and the second mirror **121** can be the same shape but of different sizes, for example, the first mirror **108** can be a larger circular shape and the second mirror **121** can be a smaller circular shape. While circular mirrors **108**, **121** are depicted, any shape mirrors can be used, for example, square, oval, and rectangular. The first mirror **108** and the second mirror **121** can have essentially the same reflectance. The first mirror **108** can have very low transmissivity (e.g., less than or equal to about 15%) to visible light and infrared light, and/or it can have essentially no observable transmissivity by a user to visible light, while the second mirror **121** can have a high transmissivity to infrared light (e.g., at least about 75% transmissivity for infrared light) and it can have very low transmissivity to visible light (e.g., at least about 80% reflectivity). The transmissivity and reflectivity of the first mirror **108** and the second mirror **121** can be similar to each other (e.g., within about 15%) for visible light but substantially different from each other for infrared right (e.g., differing by at least about 50%).

In some embodiments, the second mirror **121** can be positioned behind a window **122** of the first mirror **108**, for example, as shown in FIG. **13**. The window **122** does not extend through a thickness of the first mirror **108**. The window **122** can be a portion of the first mirror **108** that does not have the coating **110** or backing covering it. The window **122** can permit light to be transmitted through the transparent material **109** of the mirror **108** in the region of the window **122**). The window **122** can be highly transmissive (e.g., essentially transparent). When positioned in the complete mirror assembly **100**, the window **122** and the components behind it can be almost invisible to a user during use, thus hiding the appearance of a sensor assembly, discussed in more detail below.

The second mirror **121** can have a layer, backing, or coating **111** on a front surface of a transparent material **113**. The coating **111** can be the surface coupled to the coating **110** of the first mirror **108**. The coating **111** can comprise a series or stack of different and/or alternating materials, such as two or more metal oxide, silicon oxide, and/or dielectric layers, to produce a reflective optical interference effect. The color of the coating **111** can be adjusted by choosing suitable materials and/or thicknesses of the materials in the stack to match the color or appearance of the coating **110** on the rear surface of the first mirror **108**. Substantially matching the color or appearance of the coating **110** on the rear surface of the first mirror **108** and the coating **111** on the front surface of the second mirror **121** can assist in hiding (e.g., making substantially invisible) the window **122** and the components behind it from the user during use. In some embodiments the mirror **121** can include a transparent material such as glass and one or more layers. In some embodiments, the mirror **121** comprises one or more optical interference coatings or layers such as dielectric glass coatings or layers (e.g., glass with dielectric layers). In some embodiments, the mirror **121** can include an optical interference structure comprising one or more extremely thin layers (e.g., with a thickness that is less than a single wavelength of visible light) that are configured to produce an optical effect such as a predetermined reflectance and/or transmissivity of visible light at particular wavelengths or groupings of wavelengths. For example, in some embodiments, the optical interference structure can comprise a series of layers comprising at least one metallic layer and at least one dielectric layer, such as one or more alternating layers of Titanium dioxide (TiO2) and Silicon Dioxide (SiO2). The one or more alternating layers can be applied by chemical or vapor deposition.

When the rear surface of the first mirror **108** is coupled to the front surface of the second mirror **121** the coating **110** on the rear surface of the first mirror **108** and the coating **111** on the front surface of the second mirror **121** can function optically to a user's perception as one coating. For example, as shown in FIG. **14**, the assembly **126** of the first mirror **108** and the second mirror **121** can comprise a first glass layer, alternating layers of TiO2 and SiO2, metal (e.g., aluminum), and a second glass layer. The assembly **126** can include a layer of SiO2 on each side of a metal layer. A protective or opaque paint, coating, or layer can be included directly on the glass or on a metal layer and/or can be layered on top of a SiO2 layer or any other layer or structure of the assembly **126**. As used in this specification, the term "glass" is used in accordance with its ordinary meaning, which includes a silica-based hard, brittle, transparent material, typically formed in thin layers or panes. With each occurrence of "glass," it is also contemplated that non-glass substances or alternatives that are structured in a manner similar to glass or that perform in substantially the same way as glass can be used, including one or more transparent coated or uncoated polymer sheet materials such as those comprising polycarbonate or acrylic. As with all disclosure in this application, any layer or substrate or combination of layers or substrates that are disclosed in connection with this embodiment can be replaced with or combined with any other layer or substrate or combination of layers or substrates that are disclosed elsewhere herein, or omitted.

In some embodiments, the mirror assembly **100** in one region can have a first layer **109**, a second layer **110**, a third layer **111**, and a fourth layer **113**. The material(s) of which each layer **109**, **110**, **111**, **113** are made can comprise glass, metal (e.g., aluminum, silver, titanium dioxide, etc.), and a dielectric material (e.g., SiO2, etc.), and/or any other suitable material. In some embodiments, the mirror **108** can be assembled using a sputtering process, a vaporizing process, a deposition process, and/or any other suitable process. In some embodiments, one or more of the layers can be applied using adhesive, thermal bonding, sonic welding, or in any other suitable way. In some embodiments, the first layer **109** comprises glass. This can help prevent the mirror **108** from becoming scratched because glass is highly scratch resistant, much more scratch resistant than many other materials. In some embodiments, the second layer **110**, the layer disposed between the first layer **109** and the third layer **111**, comprises a metal (e.g., aluminum). In some embodiments, the third layer **111** comprises a metallic or dielectric material (e.g., TiO2, SiO2). In some embodiments, the fourth layer **113** comprises glass. One or more of the layers **109**, **110**, **111** can be used in the mirror **121**. One or more of the layers **109**, **110**, **111**, **113** can be omitted from the assembly **126**, rearranged or assembled in a different order, and/or supplemented by other layers or materials. For example, in some embodiments, a dark coating such as black paint and/or a copper layer is disposed on at least a portion of a metal layer of the mirror **108**. Any layer or structure disclosed in this specification is not required to be homogeneous, but can include multiple sub-layers or sub-structures as appropriate.

In some embodiments, an optical interference structure in the mirror **108** can include a combination of layers such as the following:

Example 1

Layer	Material	Optical Thickness	Physical Thickness (nm)
Substrate	protective paint		
9	Al	0.48	120
8	SiO2	1.0768	72.93
7	TiO2	1.2789	53.33
6	SiO2	1.2948	87.7
5	TiO2	1.3201	55.04
4	SiO2	2.3774	161.03
3	TiO2	1.3064	54.47
2	SiO2	2.1806	147.69
1	TiO2	2.0414	85.12
Medium	glass		
Total Thickness		13.3564	837.31

Example 2

Layer	Material	Optical Thickness	Physical Thickness (nm)
Substrate	protective paint		
10	SiO_2	0.7382	50
9	Al	0.48	120
8	SiO_2	1.0768	72.93
7	TiO_2	1.2789	53.33
6	SiO_2	1.2948	87.7
5	TiO_2	1.3201	55.04
4	SiO_2	2.3774	161.03
3	TiO_2	1.3064	54.47
2	SiO_2	2.1806	147.69
1	TiO_2	2.0414	85.12
Medium	glass		
Total Thickness		14.0946	887.31

Of course, many other types and orderings of layers can be used instead of or in addition to those provided in this example. In some embodiments, a reflectance of a mirror with an optical interference structure on a glass substrate can be provided as shown in FIG. **17**.

For example, as shown in FIG. **17**, for a large majority of the wavelengths of visible light, the reflectance can be at least about 95% or at least about 93% or at least about 90%. Of course, any other suitable component or series of components can be configured to provide many other or different values in a useful reflectance profile in accordance with the principles and structures disclosed in this specification.

The relative sizes of the coating layers **110**, **111** as illustrated in FIG. **14** is exaggerated as compared to the transparent material **109**, **113**, for case of illustration. The coating layers **110**, **111** can be significantly thinner than as shown as compared to the transparent materials **109**, **113**.

As shown in FIG. **14**, the first mirror **108** can be configured to permit visible light to pass through the front transparent material **109**, reflect off of the rear reflective coating or backing **110**, pass through the front transparent material **109** again, and then enter the eye of a user, forming a reflected image of the user. The combination of the first mirror **108** and the second mirror **121** can be configured to permit visible light impinging on the region of the first mirror **108** in front of the window **122** to pass through the front transparent material **109**, enter the window **122** (which can be filled with optical adhesive **114**), pass through the optical adhesive layer **114**, reflect off of the front reflective interference coating **111** of the second mirror **121**, pass through the optical adhesive layer **114** and the window **122** again, pass through the transparent material **109** again, and then enter the eye of a user, forming a reflected image of the user. The mirror assembly **100** can be configured to effectively hide the window **122** and the components behind it from the user by permitting visible light to be reflected in the region of the window **122** in essentially the same manner and at about the same reflectivity as in the regions of the first mirror **108** that are adjacent to and/or surrounding the window **122**.

Simultaneously, the transmitter **141** of the sensor **140** can emit sensing energy, such as electromagnetic energy in the infrared range, through the transparent material **113** of the second mirror **121**, through the coating **111**, through the optical adhesive, through the window **112**, and through the transparent material **109** of the first mirror **109** without substantial refraction, reflection, deviation, attenuation, and/or modification. As shown, the emitted sensing energy can reflect off of the user and then return through the transparent material **109** of the first mirror **108**, window **122**, optical adhesive **114**, coating **111**, and transparent material **113** of the second mirror **121**, and then proceed into the receiver **142** of the sensor **140**, again without substantial refraction, reflection, deviation, attenuation, and/or modification. The sensing energy is not reflected in a significant way by the coating **111** because the interference effects of the coating **111** are calculated to produce reflection for only certain wavelengths of light within the visible range. For example, the coating **111** can be configured to reflect impinging visible light in the same way as the coating **110** of the first mirror **108** so that the reflected visible light in the region of the window **122** is the same as in the other regions of the first mirror **108**.

FIG. **15** illustrates a front view of the mirror assembly **100** with the mirror **108** removed for illustrative purposes and FIG. **16** illustrates a rear view of the mirror assembly **100** with a rear portion of the housing **104** removed for illustrative purposes. The mirror assembly **100** can include a sensor assembly **140**. The sensor assembly **140** can be positioned between a rear surface of the mirror **108** and a rear surface of the housing **104**. The sensor assembly **140** can be enclosed by the mirror **108** and the housing **104**. The sensor assembly **140** can be hidden such that it is not visible by a user. In some embodiments, the sensor assembly **140** can be positioned generally central within the housing **104**. The sensor assembly **140** can include a sensor **144**. In some embodiments, the sensor **144** can be a proximity sensor or a reflective-type sensor. For example, the sensor **144** can be triggered when an object (e.g., a body part) is moved into, and/or produces movement within, a sensing region.

The sensor assembly **140** can include a power source **148** (e.g., a battery, a rechargeable battery, or a cord to be plug into an electrical outlet). The power source **148** can be positioned within the housing **104**. The power source **148** can deliver power to the light source **136**. In some embodiments, the mirror assembly **100** can include one or more weights **145** (labeled in FIG. **8**) disposed in the stand **120** of the mirror assembly **100** to counteract the weight of the sensor assembly **140**.

The sensor assembly **140** can include a transmitter **141** and a receiver **142**. The transmitter **141** can be an emitting portion (e.g., electromagnetic energy such as infrared light), and the receiver **142** can be a receiving portion (e.g., electromagnetic energy such as infrared light). The beam of light emitting from the transmitter **141** can define a sensing region. In certain variants, the transmitter **141** can emit other types of energy, such as sound waves, radio waves, or any other signals. The transmitter **141** and receiver **142** can be integrated into the same sensor or configured as separate components. The transmitter **141** and the receiver **142** can be positioned behind a rear surface of the second mirror **121**. The transmitter **141** and the receiver **142** can be aligned with and/or positioned behind the window **122** of the first mirror **108**.

In some embodiments, the mirror assembly **104** can include an LED light **143**. The LED light **143** can be positioned behind a rear surface of the second mirror **121**. The LED light **143** can be aligned with and/or positioned behind the window **122** of the first mirror **108**. The LED light **143** can be configured to emit a light in order to alert a user to a predetermined type of information. In some embodiments, the light can be emitted to indicate in one or more different ways (e.g., changing color, flashing, emitting in a steady state, etc.) that the battery of the mirror assembly

100 is charging, finished charging, and/or nearing depletion. In some embodiments, the light can be emitted to indicate that a user's presence has been identified.

In some embodiments, the sensor assembly **140** can detect an object within a sensing region. In certain embodiments, the sensing region can have a range from at least about 0 degrees to less than or equal to about 45 degrees downward relative to an axis extending from the sensor assembly **28**, and/or relative to a line extending generally perpendicular to a front surface of the sensor assembly, and/or relative to a line extending generally perpendicular to the front face of the mirror and generally outwardly toward the user from the top of the mirror assembly. In certain embodiments, the sensing region can have a range from at least about 0 degrees to less than or equal to about 25 degrees downward relative to any of these axes or lines. In certain embodiments, the sensing region can have a range from at least about 0 degrees to less than or equal to about 15 degrees downward relative to any of these axes or lines.

If the receiving portion detects reflected energy (e.g., above a threshold level) from an object within the beam of light emitted from the transmitter **141**, the sensor assembly **140** sends a signal to a controller to activate the light source **136**. Once the light source **136** activates, the light source **136** can remain activated so long as the sensor assembly **140** detects an object in a sensing region. Alternatively, the light source **136** can remain activated for a pre-determined period of time. For example, activating the light source **136** can initialize a timer. If the sensor assembly **140** does not detect an object before the timer runs out, then the light source **136** is deactivated. If the sensor assembly **140** detects an object before the timer runs out, then the controller can reinitialize the timer, either immediately or after the time runs out.

Conditional language, such as "can," "could," "might," or "may," unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

The terms "comprising," "including," "having," and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term "or" is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term "or" means one, some, or all of the elements in the list.

The ranges disclosed herein also encompass any and all overlap, sub-ranges, and combinations thereof. Language such as "up to," "at least," "greater than," "less than," "between" and the like includes the number recited. Numbers preceded by a term such as "about" or "approximately" include the recited numbers. For example, "about 5 mm" includes "5 mm."

The term "horizontal" as used herein is defined as a plane parallel to the ground or floor on which the component or device is positioned in normal use, either directly or indirectly. The term "vertical" refers to a direction perpendicular to the horizontal as just defined. Terms such as "above," "below," "bottom," "top," "side," "higher," "lower," "upper," "over," and "under," are defined with respect to the horizontal plane.

Although certain embodiments and examples have been described herein, it will be understood by those skilled in the art that many aspects of the devices shown and described in the present disclosure may be differently combined and/or modified to form still further embodiments or acceptable examples. All such modifications and variations are intended to be included herein within the scope of this disclosure. A wide variety of designs and approaches are contemplated. No feature, structure, or step disclosed herein is essential or indispensable.

For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

Moreover, while illustrative embodiments have been described herein, the scope of any and all embodiments having equivalent elements, modifications, omissions, combinations (e.g., of aspects across various embodiments), adaptations and/or alterations as would be appreciated by those in the art based on the present disclosure. The limitations in the claims are to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive. Further, the actions of the disclosed processes and methods may be modified in any manner, including by reordering actions and/or inserting additional actions and/or deleting actions. It is intended, therefore, that the specification and examples be considered as illustrative only, with a true scope and spirit being indicated by the claims and their full scope of equivalents.

The following is claimed:

1. A mirror assembly comprising:

a first mirror with a front surface and a rear surface, the front surface visible by a user;

a second mirror having a front surface and a rear surface, the front surface of the second mirror coupled to the rear surface of the first mirror;

a light source disposed on or at least partially around the first mirror; and

a sensor assembly positioned behind a rear surface of the second mirror, the sensor assembly comprising:

a transmitter positioned behind the rear surface of the second mirror; and

a receiver positioned behind the rear surface of the second mirror;

wherein the light source is configured to be activated by the sensor assembly.

2. The mirror assembly of claim **1**, wherein the sensor assembly comprises a proximity sensor.

3. The mirror assembly of claim **1**, wherein the transmitter and the receiver are positioned behind a window of the first mirror.

4. The mirror assembly of claim **3**, further comprising an LED light positioned behind the window of the first mirror, wherein the LED light can be configured to emit a light when the mirror assembly is charging.

5. The mirror assembly of claim **3**, wherein the window is substantially invisible to the user.

6. The mirror assembly of claim **1**, wherein the second mirror is smaller than the first mirror.

7. The mirror assembly of claim **1**, wherein the first mirror and the second mirror have the same curvature.

8. The mirror assembly of claim **1**, wherein the second mirror is coupled to the rear surface of the first mirror via an adhesive.

9. The mirror assembly of claim **8**, wherein the index of refraction of the adhesive is substantially the same as the index of refraction of a transparent material of the first mirror.

10. The mirror assembly of claim **1**, wherein the first mirror has a coating on the rear surface and the second mirror has a coating on the front surface.

11. The mirror assembly of claim **10**, wherein the coating of the first mirror is aluminum and the coating of the second mirror is an optical interference coating.

12. The mirror assembly of claim **1**, further comprising a power source positioned within a housing of the mirror assembly.

13. The mirror assembly of claim **1**, further comprising a shaft coupled at a first end to a housing of the mirror assembly and at a second end to a stand.

14. The mirror assembly of claim **13**, wherein the shaft is rotatably coupled at the first end to the housing and the shaft is rotatably coupled at the second end to the stand.

15. A mirror assembly comprising:

a first mirror with a front surface and a rear surface, the front surface visible by a user;

a second mirror having a front surface and a rear surface, the front surface of the second mirror coupled to the rear surface of the first mirror; and

a sensor assembly positioned behind the rear surface of the second mirror, the sensor assembly comprising:

a transmitter configured to emit light; and

a receiver configured to receive light;

wherein the sensor assembly is configured to detect a user's presence.

16. The mirror assembly of claim **15**, wherein the sensor assembly comprises a proximity sensor.

17. The mirror assembly of claim **15**, further comprising a light source, wherein when a user's presence is detected the light source is activated.

18. The mirror assembly of claim **15**, wherein the transmitter and the receiver are positioned behind a window of the first mirror.

19. The mirror assembly of claim **15**, wherein the second mirror is smaller than the first mirror.

20. The mirror assembly of claim **15**, wherein the first mirror and the second mirror have the same curvature.

21. The mirror assembly of claim **15**, wherein the second mirror is coupled to the rear surface of the first mirror via an adhesive.

22. The mirror assembly of claim **15**, wherein the first mirror has a coating on the rear surface and the second mirror has a coating on the front surface.

23. The mirror assembly of claim **22**, wherein the coating of the first mirror is aluminum and the coating of the second mirror comprises a dielectric layer.

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